

NON-INTERCHANGEABLE SCORING PROTOCOLS AND A DRAW-COUNT-MATCHED REFERENCE: A NARROW MEASUREMENT STUDY ON LLaDA-8B AND GSM8K-MC

Anonymous authors

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ABSTRACT

Multiple-choice accuracy for a diffusion language model is not determined by the model alone: the scorer chooses which option tokens are visible, when they are revealed, and whether revealed ground-truth tokens are fed back into later steps. We audit that measurement construction on one bounded bed: LLaDA-8B-Instruct with balanced four-choice GSM8K-MC.

On a confirmatory $n=400$ pool `s_conf` exceeds the non-injected `mc_uniform` reference by +13.0 pp at the per-candidate $n_{mc,i} = 2L_i$ total-NFE-fair equalization point ($b/c = 109/57$, exact $p = 6.6 \times 10^{-5}$; four tested budgets, conservative read +9.0 pp). This *refutes* the narrow account that the advantage is an artifact of the reference receiving less compute, but establishes no budget-independent law. On a disjoint fresh pool a fixed left-to-right order whose *schedule selection* reads no model output still beats `s_conf`; confidence ordering is therefore neither necessary nor uniquely optimal on this answer format. Rank-agreement and distractor-perturbation diagnostics support only a bed-local, sign-consistent non-interchangeability reading; we do not claim dose-response, cross-bed transfer, or isolated causal attribution to feedback.

1 INTRODUCTION

Masked diffusion language models assign a multiple-choice answer only after a scoring protocol turns option-level denoising losses into a ranking. That protocol is part of the measurement instrument. In particular, a scorer may reveal candidate tokens sequentially and feed the known candidate token back into the context, or it may score a randomly masked subset without such feedback. Even two feedback-based scorers can use different reveal schedules. Treating their accuracies as if they were readings from the same ruler hides a change in what was measured.

We report a deliberately narrow audit of that construction. The evidence is one LLaDA-8B-Instruct $\times$ GSM8K-MC four-option construction evaluated on two historical balanced pilot pools, a confirmatory 400-question pool for the reference-arm contrast, and a disjoint fresh 819-question pool for reveal order. The study makes three claims, each tied to its pool and boundary (Table 3, Table 6). *First*, on the tested $n=400$ reference-budget curve, `s_conf`’s advantage over the non-injected reference is positive and significant at every tested budget and narrows as the reference is given more compute (Table 3, Figure 2). The registered matched- L point remains the primary endpoint and is measured on that bed; the per-candidate total-NFE-fair equalization at $2L$ (the decisive read, both arms $2L_i$ forwards) is measured on the same bed and, within this $n=400$ endpoint change, reports the same headline +13.0 pp gap (exact $p = 4.8 \times 10^{-5}$ and 6.6×10^{-5}); under the paper’s refusals this *does not* license an estimand-external claim that endpoint choice is irrelevant in general, only that on this tested bed the swap leaves the headline unchanged. The highest-budget read remains the conservative one. This *refutes* a less-reference-compute-only account, not a budget-independent one, and it is a complete protocol swap, not causal isolation. *Second*, reveal *order* on injected prompts, measured against the *tested persisted schedule*, has a surviving effect on the disjoint fresh pool under its $k=1$ multiplicity family; but a trivial fixed order whose *schedule selection* reads no model output still beats `s_conf` on that pool on average (Table 6) — a margin that strengthens in the answer-pool

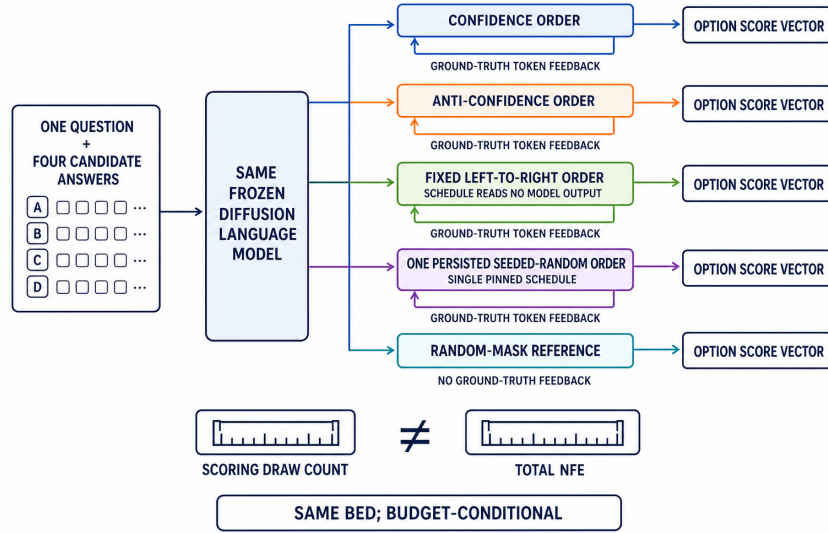


Figure 1: Conceptual scoring-protocol fork, and the measurement-level view of this study. The same item and model feed all five arms of Table 1: confidence, anti-confidence, fixed, seeded-random, and random-mask paths. Feedback arrows stay inside the four injected paths; the random-mask path has none. Each path ends in the same generic score-vector object, so no winner or item-level ranking is drawn, and the two rulers distinguish nominal draw count from total NFE. This generated artwork contains no experimental values; all quantitative results are data-derived.

construction stratum and *reverses* at answer position 0 (Table 7). Confidence ordering is therefore not necessary, not uniquely optimal, and not exclusively causal; a non-random structure—including answer format—can suffice. *Third*, protocols are not interchangeable: changing the scorer changes per-question option rankings and choice-set sensitivity on this bed. The pairwise cross-protocol Kendall τ sits only just above the random arm’s *own* self- τ floor, so it bounds protocol-specific disagreement rather than proving distinctness; the deterministic `s_conf-fixed_inject` rank agreement is separately consistent with genuinely distinct protocols (Table 15).

The contribution is a protocol audit rather than a new scoring algorithm: a paired, decision-record-level decomposition of comparisons that would otherwise collapse model and scorer into one accuracy number, with the negative boundary stated up front—neither the design nor its measurements support a dose-response curve, a concavity claim, or transfer to another model, dataset, or judge.

2 RELATED WORK AND LOCAL EVIDENCE CONTEXT

LLaDA introduced the 8B masked-diffusion model used here (Nie et al., 2025); iLLaDA later introduced confidence-based multiple-choice scoring (Nie et al., 2026). That scorer reveals the most confident candidate token and feeds the known token into later steps, so a comparison with a Monte Carlo-style scorer replaces protocol components as well as the accounting rule. We audit that replacement, not the published benchmark values. GSM8K (Cobbe et al., 2021) supplies only the controlled four-choice bed; our claims do not transfer beyond it.

Scorer dependence is established more broadly: benchmarks vary with task and prompt format (Brown et al., 2020; Hendrycks et al., 2021; Lin et al., 2022; Chen et al., 2021); frameworks distinguish raw from length-normalized scoring and expose prompt sensitivity (Liang et al., 2023; Biderman et al., 2024; Zhu et al., 2024); and audits cover judge design, contamination, and aggregation (Perez et al., 2022; Jain et al., 2024; Wang et al., 2023). Our contribution is the measured masked-diffusion protocol swing on a stated budget curve.

The closest mechanism literatures are mask-out objectives (Nie et al., 2025; 2026; Sahoo et al., 2024), multiple-choice normalization and position sensitivity (Holtzman et al., 2021; Zheng et al., 2024), and confidence-guided diffusion decoding (Nie et al., 2025). Our controls extend these con-

Table 1: The five scoring arms. L_i is candidate length. The scoring-forward count excludes the additional L_i adaptive schedule-selection forwards for confidence and anti-confidence order; nominal draw matching is therefore not total-NFE matching.

Arm	Reveal schedule	GT feedback	Scoring forwards
s_conf	min-CE first	yes	L_i (+ L_i select)
rand_inject	seeded random	yes	L_i
anti_inject	max-CE first	yes	L_i (+ L_i select)
fixed_inject	fixed order	yes	L_i
mc_uniform	random mask	no	1 ($n_{mc}=1$)

cerns to token injection: a fixed left-to-right schedule can beat confidence ordering on this answer format, but no cited work or result here establishes a transferable injected-feedback effect.

3 SETUP

Bed and pairing. Each item pairs one GSM8K question with four candidate answers scored by LLaDA-8B-Instruct (Nie et al., 2025) with frozen weights. Choice sets were frozen but differ by pool, and the fresh pool is *internally* heterogeneous by construction: the historical, $n = 400$, and first 336 fresh items use deterministic near-miss distractors; the remaining 483 fresh items use three distinct values drawn from a frozen GSM8K answer pool. Every arm reuses the recorded choice set. Historical and $n = 400$ pools are exactly answer-position balanced; the fresh pool is approximate. Fresh-pool contrasts therefore mix protocol effects with construction stratum, and post-hoc position standardization leaves the fresh-pilot difference, so we report them as strata-confounded. Inference uses BF16, no temperature, and no generation. Confidence and fixed schedules are deterministic; the seeded-random result is one persisted schedule. Because its runner used an unpinned Python string hash, replay is anchored to stored per-candidate orders, not the nominal seed alone. Appendix H lists identifiers, prompts, orders, and seeds. Historical pools use 12 or 16 items per answer position ($n = 48/64$); the $n = 400$ and fresh $n = 819$ pools are non-overlapping. Within each pool, arms score the same items, so contrasts are paired by question. Top-1 exact correctness is the sole endpoint; no partial credit is assigned.

Five arms. For candidate tokens $y_{1:L_i}$, an injected arm chooses a permutation of positions, reads cross-entropy before each reveal, inserts the known candidate token at that position, and repeats until the candidate is visible. It sums the per-step cross-entropies and selects the option with minimum score. s_conf chooses the next minimum-cross-entropy position; rand_inject uses a deterministically seeded random order; anti_inject chooses the maximum-cross-entropy position; and fixed_inject uses fixed left-to-right order. The fifth arm, mc_uniform, draws a masked subset of size $k \sim U\{1, \dots, L\}$ at each of n_{mc} draws (LLaDA upper-bound estimator), reads the partial-mask cross-entropy at the masked positions, and rescales per option by the standard factor L_i/k , so each draw is an unbiased length-corrected estimate of the full mask-out score and the arm returns their mean; ties break by first index. The $n_{mc}=1$ single draw is this paper’s minimum-budget arm; it differs operationally from the LLaDA/iLLaDA default confidence scorer, whose reveal targets the model’s own argmax token rather than the injected ground-truth candidate. Thus “protocol family” denotes the full tuple of feedback rule, reveal schedule, and scoring dose—not the model name alone.¹ Table 1 summarizes these protocol components.

Statistics and derived diagnostics. For paired binary correctness indicators, b and c count discordant directions, $m = b + c$, and $\hat{\Delta} = (b - c)/n$. We report the exact two-sided McNemar test (McNemar, 1947) and the locked 95% interval. Paired intervals default to the registered standard error $\widehat{SE} = \sqrt{m/n}$; each question-bootstrap or paired-bootstrap interval is named as such in its own caption. An alternative mean-corrected form moves the reported registered-SE bounds negligibly on the $n = 400$ pool and is convention-sensitive only on the small pilot pools, changing no verdict anywhere—Appendix Table 10 prints both conventions side by side. The pre-registered interpretation gate requires $m \geq 25$ before a paired contrast is promoted. For protocol agreement,

¹The anonymous artifact describes the exact sampler, prompt, mask construction, random-seed policy, and first-index tie rule.

Kendall τ (Kendall, 1938) is recomputed within each item from the two four-option score vectors after dropping tied pairs, then averaged across items. Choice-set sensitivity is an offline deterministic overlay: each of the three distractors is shrunk by 25%, and a flip is counted if the selected option changes (with perturbation counts stated per pool in the appendix). These are descriptive protocol diagnostics, not hypothesis tests.

Load-bearing estimand. The primary announcement is the paired mean accuracy gap $\Delta = \mathbb{E}_i[\mathbb{I}(A(q_i)=y_i) - \mathbb{I}(B(q_i)=y_i)] = \Pr(A(q)=y) - \Pr(B(q)=y)$ under nominal draw-count matching: q_i is a fixed item tuple (question, designated answer, deterministic four-option pool from the same scorer), the expectation is over the n items of a named pool, and the budget unit is the nominal per-candidate scoring draw (each arm spends L_i forwards per candidate; nominal, not total NFE, per Table 1). Estimation is the paired $\hat{\Delta} = (b - c)/n$ with the exact two-sided McNemar test; Δ is a protocol gap, not a mechanism attribution. Registration kept the nominal matched- L point ($n_{mc}=L$, draw-count only) primary; that primary point and the per-candidate total-NFE-fair equalization at $2L$ are both measured on the $n=400$ bed, and the reported curve still treats the +9.0 pp high-budget endpoint as its conservative read (Table 3).

Multiplicity. The confirmatory five-arm analysis uses all ten pairwise tests as one Holm family (Holm, 1979) at $\alpha = 0.05$; the smallest-step threshold is 0.005. The fresh fixed-order adjudication has one preregistered load-bearing pair (`s_conf-fixed_inject`, $k = 1$). As a conservative sensitivity, we also name a five-test reporting family: `s_conf` versus `rand_inject`, `anti_inject`, `fixed_inject`, `single-draw mc_uniform`, and `draw-count-matched mc_uniform`. Under Holm at $\alpha = 0.05$, the fixed-order result remains significant even under the worst-case factor-five adjustment ($p = 0.0194$). This named five-test set is a robustness disclosure, not a retroactive change to the registered $k = 1$ endpoint.

Registration and principal-contrast selection. The initial registration fixes the bed, five arms, $m \geq 25$ promotion gate, and exact paired testing. Draw-count matching was selected only after the small-pool scan, so every small-sample row is historical/exploratory. A separate confirmatory registration fixed the $n = 400$ draw-count-matched contrast, the ten-test Holm family, and the power stop before data. Its planning MDE and its magnitude gate are distinct, unchanged quantities: the planning MDE sized the $n = 400$ pool, while the same 4 pp promotion bar — a *point-estimate* gate, not an inferential lower-bound gate, so the high-budget read’s lower bound of 3.46 pp neither clears 4 pp nor needs to — applies to the matched- L , total-NFE-fair $2L$, and high-budget reads on that bed, all three of which clear it (Table 3). A separately pre-registered exploratory 25 pp tier applies only to the historical small rows, not as a stricter second bar for any $n = 400$ read; this unified rule *refutes* the earlier asymmetric thresholding. The corrected high-budget $n = 400$ run and the anonymous decision chronology are in Appendix Tables 14 and 13.

4 THE REFERENCE REMAINS LOWER ACROSS THE TESTED BUDGET CURVE ON THIS BED

The answer-level budget curve joins independently registered $n = 400$ runs, with $n_{mc}=128$ as the conservative high-budget main read. Its decisive row is the total-NFE-fair $2L$ point, where both arms spend $2L_i$ forwards per candidate (Table 3). The pre-registered nominal matched- L endpoint instead set $n_{mc,i} = L_i$, matching only the injected arm’s L_i scoring steps and not its schedule-selection passes, token-forwards, or total NFE; that draw-count-matched point is separately measured on this bed (paired $b/c = 106/54$), and the $2L/2L$ row returns the same +13.0 pp gap from a numerically distinct paired record ($b/c = 109/57$), so the two endpoints differ in discordance pattern but not in headline effect. The candidate-length distribution is in Appendix Table 16. At the headline operating point the paired result survives the ten-test Holm family; Table 3 and Figure 2 report the arm accuracies, discordances, estimate, interval, and exact test.

Same-bed likelihood readout. For each deterministic injected policy, summed candidate cross-entropy is the exact negative log-likelihood under that sequential unmasking policy. Since selection is its arg min, score and likelihood rankings coincide *by construction*; this identity is not an additional comparison. Table 2 shows both ground-truth signal and a policy shift on the same questions:

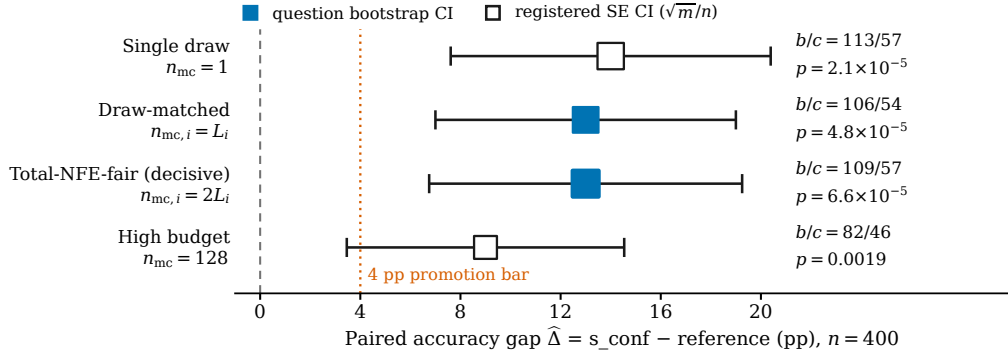


Figure 2: Data-derived paired forest of the tested $n=400$ reference-budget curve: the same four contrasts, frozen records, and intervals as Table 3, with no new decoding. Square area scales with the discordant count m ; filled squares mark question-bootstrap intervals and open squares registered paired-SE intervals. The matched- L primary endpoint and the decisive total-NFE-fair $2L$ equalization coincide at +13.0 pp, while the reference improves with budget, contracting the gap to the conservative +9.0 pp read. All four intervals exclude zero. The dotted 4 pp line is the registered *point-estimate* promotion bar, not a lower-bound requirement, so the high-budget interval reaching 3.46 pp is expected (§3). Regenerated by `paper/src/figures/decisive_paired_forest_refine10.py`.

Table 2: Same-bed per-token negative log-likelihood (nats; lower is better; $n=400$). ρ relates true-option nats to correctness. For `fixed_inject/s_conf`, paired true-option directions are 349/40 with 11 ties ($n_{\text{selected}}=389$, exact two-sided $p = 1.1 \times 10^{-62}$). `mc_uniform` is the separate mask-out family.

Arm	chosen nats/tok	true-opt nats/tok	ρ (true-opt, correct)	acc. %
<code>s_conf</code>	1.75	1.88	-0.12	42.75
<code>fixed_inject</code>	1.02	1.21	-0.28	45.50
<code>anti_inject</code>	1.20	1.43	-0.35	40.75
<code>rand_inject</code>	1.18	1.62	-0.41	39.00
<code>mc_uniform</code> (ELBO family)	0.19	0.77	-0.43	28.75

true-answer readout falls from 1.88 to 1.21 nats/token under `s_conf` versus fixed order. Thus the gap is localized to a protocol-dependent likelihood readout, without a cross-bed claim. These are teacher-forced option likelihoods, not generation; `mc_uniform` remains a separate mask-out family whose ELBO-style estimand is not point-comparable to sequential reveal.

These are operating points of one replacement, not independent triangulation. The monotone reference rise contracts but does not erase the gap, refuting a pure low-compute artifact on the tested curve, exact $2L$ included; any budget-independent law off the measured points remains untested, and the paired result isolates neither feedback, schedule, nor budget.

Table 4 shows every tested reference above its pool’s chance level at the nominal one-sided level and farther above it with budget; under Bonferroni-5 the two lowest-budget $n=400$ reads become *undetermined* against chance rather than at chance, which does not touch the paired contrasts in Table 3 and Appendix Table 14. The $n = 400$ high-budget read is significant and clears the same 4 pp bar as the primary, whereas the $n = 64$ pilot remains underpowered under its exploratory tier and is superseded, at the identical $n_{mc}=128$ operating point, by that measured $n=400$ row. Thus chance calibration and compute both *refute* a pure budget artifact; budget independence as a law off the tested curve stays *honestly unresolved*. The rest of the nested $n = 64$ scan is retained only as

³The $n=400$ single-draw row uses `results/c1_n400_20260819-104302/perq_r0.jsonl`; the matched- L row uses `results/nmc_sweep_grid128_20260823-023717/perq_r0.jsonl`; the $2L/2L$ (total-NFE-fair) and high-budget rows use `results/nmc_sweep_grid128_20260824-192820/perq_r0.jsonl` ($n_{\text{selected}}=400$ in each). “Same bed” is a checked property of the two records, not an assumption: `s_conf` is per-item identical across them (171/400) and so is $n_{mc}=128$ (135/400); only the two reference samplers differ.

Table 3: Tested reference-budget curve in ascending reference budget (reference/s_{conf} total NFE in parentheses; row labels are the per-candidate budget labels of Appendix Table 17). The decisive row is total-NFE-fair equalization at $2L$, where the reference uses $n_{mc,i} = 2L_i$ forwards and s_{conf} L_i scoring plus L_i schedule-selection forwards. The pre-registered nominal matched- L point (reference $n_{mc,i} = L_i$, nominal draw-count match only, primary endpoint per registration) is measured on the same $n=400$ items as that decisive row, with the same s_{conf} trajectories (171/400) and the same +13.0 pp headline, so the two endpoints are numerically distinct paired records with interchangeable headline readings.³ The two $n_{mc,i}$ rows quote question-bootstrap intervals (20k, seed 20260824) and the rest the registered paired SE $\sqrt{m/n}$; tests are exact two-sided McNemar; Figure 2 plots the four $n=400$ rows. The $n=64$ scan is exploratory and superseded.

Reference point (ref/s _{conf} NFE)	n	Acc. pair (s _{conf} /ref)	b/c	$\hat{\Delta}$	95% CI	Exact p	Role
Single draw (1/2L)	400	42.75/28.75%	113/57	+14.0 pp	[7.61, 20.39]	2.1×10^{-5}	corroboration
$n_{mc,i} = L_i$ (L/2L)	400	42.75/29.75%	106/54	+13.0 pp	[7.00, 19.00]	4.8×10^{-5}	pre-registered primary
$n_{mc,i} = 2L_i$ (2L/2L)	400	42.75/29.75%	109/57	+13.0 pp	[6.75, 19.25]	6.6×10^{-5}	NFE-fair equalization (decisive)
$n_{mc} = 128$ (128/2L)	400	42.75/33.75%	82/46	+9.0 pp	[3.46, 14.54]	0.0019	conservative read
$n_{mc,i} = L_i$ (L/2L)	64	53.12/28.12%	24/8	+25.0 pp	[7.68, 42.32]	0.007	exploratory (superseded)

Table 4: Reference-arm calibration against the chance level of each declared four-choice pool (exact one-sided binomial tests on the arm’s own correct count). Rows 2–4 are the same $n = 400$ pool at three scoring budgets; the last row is a historical pilot on a different pool, not a large-pool substitute. These five tests are the paper’s one uncorrected family, so the last column scores each against the Bonferroni-5 bar $\alpha = 0.05/5 = 0.01$: three rows clear it and the two lowest-budget $n=400$ reads do not. The per-candidate draw-matched row sets $n_{mc,i}=L_i$ per candidate. Those two remain corroborative chance checks either way; no headline claim rests on this family, because the headline claims are the paired McNemar contrasts of Table 3, which are unchanged by any correction applied here.

Reference arm	n	Correct	Accuracy	Chance	Exact p	Bonf.-5
Seeded-random reveal, fresh	819	398	48.60%	25%	6.56×10^{-48}	clears
mc_uniform, single draw	400	115	28.75%	25%	0.0486	no
mc_uniform, per-candidate draw-matched	400	119	29.75%	25%	0.0176	no
mc_uniform, $n_{mc} = 128$	400	135	33.75%	25%	5.6×10^{-5}	clears
mc_uniform, $n_{mc} = 128$ pilot	64	27	42.19%	25%	0.0019	clears

exploratory evidence (Appendix Table 14, Figure 5), and the identification projection in §7 is not answer accuracy.

Analysis map. Table 5 consolidates every headline estimand: pool, registration status, direction, multiplicity family, primary interval convention, and final verdict. It is the single index into the Tables 3, 4, 6, 7 and 15 evidence trail; per-estimand alternative conventions remain in Appendix Tables 14, 10.

5 SCORING PROTOCOLS ARE NOT INTERCHANGEABLE ON THIS BED

The score vectors produced by s_{conf} and fixed_inject—two deterministic protocols, so no schedule-seed noise enters—disagree clearly in rank ($\tau \approx 0.48$; Table 15), consistent with genuinely distinct protocols. Between s_{conf} and the seeded-random rand_inject, the item-level τ means sit only 0.01–0.10 above the random arm’s own self- τ floor (Figure 3); because that floor’s dispersion exceeds the gap, we read that value as an *upper bound* on protocol-specific rank disagreement rather than as evidence of distinctness, and the deterministic pair carries the load-bearing

Table 5: Analysis map: headline estimands of this paper. *Pool* = question bed; *Prereg. status* = whether the endpoint was fixed before observing data; *Direction* = pre-registered read direction (+ = first-listed arm better, − = second-listed arm better, 0 = symmetric/diagnostic); *Family* = multiplicity correction under which the displayed *p* is read; *Primary interval* = the registered or displayed uncertainty convention; *Verdict* = the registered or post-hoc reading (Holm survive / Bonf. clear / CI or LB excludes 0 or fails to). Discounted secondary diagnostics (chance calibration, strata) follow the same per-row convention.

Estimand	Pool	Status	Dir	Family	Verdict	Table
s_conf mc_uniform matched- <i>L</i>	vs <i>n</i> =400	pre-reg. mary	pri- +	Holm-10	Holm (+13.0 pp, [7.00, 19.00])	survives 3
s_conf mc_uniform 2 <i>L</i> /2 <i>L</i>	vs <i>n</i> =400	post-sel. (decisive)	sec. +	Holm-10	Holm (+13.0 pp, [6.75, 19.25])	survives 3
s_conf mc_uniform 128/2 <i>L</i>	vs <i>n</i> =400	pre-registered	+ +	Holm-10	Holm vives (+9.0 pp, [3.46, 14.54])	sur- 3
s_conf rand_inject fresh	vs <i>n</i> =819	post-sel. sec.	+ +	<i>k</i> =1 LB)	(1s LB >0 (+12.58 pp)	6
s_conf fixed_inject fresh	vs <i>n</i> =819	pre-reg. single pair	−	<i>k</i> =1 (5-test Holm)	A: fixed (−5.62 pp)	wins 6
s_conf fixed_inject pilot	vs <i>n</i> =400	pre-reg. con- trol	−	reg. MDE ₈₀	NOT DETECTED	6
five-option probe	<i>n</i> =64	pre-reg. 25	Tier- +	discordance gate	UNDETERMINED	14 [‡]
s_conf rand_inject pos 0	vs fresh pos 0	post-hoc (Bonf. 4)	+ +	Bonferroni- 4	Bonf. (+12.56 pp)	survives 7
s_conf mc_uniform chance	vs <i>n</i> =400 draws	post-hoc 5	fam- +	Bonferroni- 5	3/5 clear ($\alpha=0.01$)	4
protocol τ	<i>n</i> =64	descriptive	0 −		distinct by floor (0.48)	15

Primary interval conventions by row (all Table 3 and remaining paired rows): registered paired SE $\sqrt{m}/n$ for matched-*L* and 128/2*L*; question bootstrap (B=20k) for 2*L*/2*L*; paired bootstrap for fresh; registered one-sided LB for fresh $\hat{\Delta}$; exact McNemar *p* for strata; exact binomial for chance. [‡]Five-option probe full counts in Appendix Table 14.



Figure 3: Data-derived per-question Kendall τ distributions for s_conf versus rand_inject. Points are deterministic question-level values; boxes show quartiles and the center line shows the median. The mean printed below each box is the locked overlay value rounded to three decimals (Table 15 gives four). The comparison floor is the random arm’s own self- τ (mean 0.2361, Table 15); the cross-protocol means sit only 0.01–0.10 above it, i.e. within the floor’s dispersion, so this panel is descriptive, not inferential.

reading. Replacing the scoring protocol is accordingly not a harmless implementation detail, even among protocols that use ground-truth feedback.

Table 6: Reveal-order evidence and attribution control. Fresh is secondary, pilot undetermined, and pooled a heterogeneous-pool disclosure; their magnitudes differ ($z=2.41$, $p=0.016$; robust $z=2.32$). The fixed control bounds `s_conf`’s advantage at +2.32 pp (95% CI upper), and only 2.1% of order traces match. Cells mark one-sided (1s, constant 2.4865) or two-sided (2s, 2.8016) intervals/MDE₈₀, each as registered per endpoint (the fresh endpoint’s registered rule is one-sided); Figure 4 uses the paired-bootstrap two-sided CI for the fresh row as a display convention. Raw exact two-sided McNemar p per row: fresh 204/101, $p=3.7 \times 10^{-9}$; fresh-fixed 99/145, $p=0.0039$; fixed control $p=0.334$ (pilot/pooled as per their registered rules). All use paired SE $\sqrt{m}/n$. The fresh row collapses the two construction strata that Table 7 reports separately.

Reading (contrast)	n	$\hat{\Delta}$	m	Interval / LB	MDE ₈₀	Role
Fresh ^d	819	+12.58 pp	305	1s LB +9.07 pp 2s CI	5.30 pp (1s) 8.32 pp	post-selection secondary
Pilot ^d	400	+3.75 pp	141	[−2.07, +9.57] pp	(2s) 4.31 pp	undetermined secondary
Pooled ^d	1219	+9.68 pp	446	1s LB +6.83 pp 2s CI	(1s) 7.24 pp (2s)	disclosure ^a
Fresh fixed ^{b,e}	819	−5.62 pp	244	[−9.36, −1.88] pp 2s CI	5.34 pp (2s) 7.24 pp (2s)	A: fixed wins not detected
Fixed control ^e	400	−2.75 pp	107	[−7.82, +2.32] pp	CI upper +2.32 pp ^c	($p=0.334$)

^a A sample-size-weighted average of the two heterogeneous pools, *not* a cross-bed mechanism constant; retained for completeness after the registered heterogeneity check ($z=2.41$, $p=0.016$) triggered the disclosed post-data elevation of the fresh endpoint.

^b Fresh `s_conf` versus `fixed_inject` is the single pre-specified pair ($k=1$) and survives the worst-case five-test Holm family ($p=0.0194$). Tagged MDE₈₀ cells use the caption’s tail convention and claim no equivalence; Appendix Table 10 shows the alternative SE convention changes no verdict.

^c That row’s role is an exclusion rather than a detection, so it intentionally carries *both* its MDE₈₀ and its CI upper bound.

^d Contrast: `s_conf` versus `rand_inject` (seeded-random). ^e Contrast: `s_conf` versus `fixed_inject` (fixed order).

Choice-set perturbation provides a second, operational reading. Figure 6 plots the exact-count rates for both protocols at $n = 64$; the historical $n = 48$ panel is sign-consistent and omitted for space. The between-protocol gap persists in sign across the two samples, although it narrows in the larger sample. The 25% overlay is a design-chosen proportional shrink of each protocol’s own distractor scores, not a scale-invariant or scale-fair one, so we compare only persistence and sign, never cross-scorer magnitudes, and no significance or power claim attaches to these deterministic perturbations. Together with rank disagreement, they support only a sign-consistent, bed-local non-interchangeability reading.

6 STRUCTURED REVEAL BEATS THE TESTED SEEDED-RANDOM POLICY, BUT CONFIDENCE IS NOT UNIQUE

Sections 4–5 establish the reference gap and bed-local non-interchangeability, not reveal order. After the registered $n=400$ contrast was *undetermined* (Appendix A), a fresh, disjoint $n=819$ pool used the same pairing; its endpoint is one recorded seeded-random policy, not a schedule-seed average.

Persisted-schedule sensitivity. The single pinned seeded-random schedule calls for a comparability audit against other schedule realizations. Two *independent events* observed on the same $n=819$ pool sharpen the reading without further decoding. First, the registered between-pool heterogeneity check (pilot $n=400$ vs. fresh $n=819$, same `s_conf` vs. `rand_inject` contrast) returns $\delta_{\text{diff}} = 8.83$ pp with $z = 2.41$, two-sided $p = 0.016$; magnitude shifts of at least ~ 8 pp therefore arise on this bed from pool composition alone, with the schedule *pinned*. The post-data choice to break the tie by elevation of the fresh endpoint is disclosed in Table 8, and the fresh row carries the “post-selection secondary” role in Table 6. Second, the order-trace match share between the persisted schedule and the fixed left-to-right order is 2.1% (17/819; Table 6 caption): the persisted schedule is a single, non-representative schedule realization, not an average over the seeded-random family. Under the registered heterogeneity observation, cross-realization magnitude shifts of magnitude at

Table 7: Fresh-pool strata split on the enlarged $n=819$ bed; b/c and p are paired discordants and exact two-sided McNemar. The three positive estimates reported for this contrast are distinct estimands, not replications of one number: fresh s_conf -vs- $rand_inject$ +12.58 pp ($n=819$), the paired-bed s_conf -vs- $mc_uniform$ +13.0 pp ($n=400$ on both the $L/2L$ and the $2L/2L$ endpoint), and the heterogeneous pooled s_conf -vs- $rand_inject$ +9.68 pp ($n=1219$).⁵ The position audit is post-hoc; see the text for its Bonferroni-4 control.

Split	vs	n	Acc. pair (s_conf /other)	$b/c; \hat{\Delta}$	exact p
Constr.: near-miss (336)	$rand_inject$	336	47.02/33.33%	90/44; +13.69 pp	8.7×10^{-5}
Constr.: answer-pool (483)	$rand_inject$	483	71.01/59.21%	114/57; +11.80 pp	1.6×10^{-5}
Constr.: near-miss (336)	$fixed_inject$	336	47.02/50.00%	51/61; -2.98 pp	0.395
Constr.: answer-pool (483)	$fixed_inject$	483	71.01/78.47%	48/84; -7.45 pp	0.0022
Answer pos. 0	$fixed_inject$	199	65.83/53.27%	46/21; +12.56 pp	0.0031
Answer pos. 1	$fixed_inject$	207	61.84/72.46%	15/37; -10.63 pp	0.0032
Answer pos. 2	$fixed_inject$	226	62.83/74.34%	18/44; -11.50 pp	0.0013
Answer pos. 3	$fixed_inject$	187	53.48/65.78%	20/43; -12.30 pp	0.0052

Reveal-order effect by pool (Delta = s_conf - $rand_inject$)

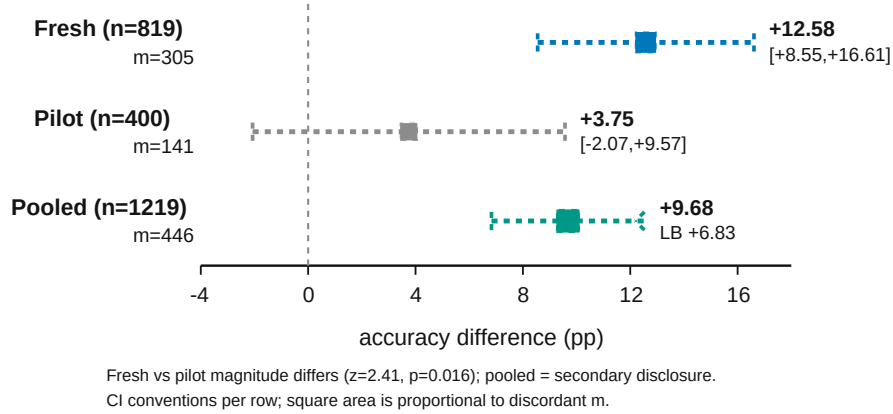


Figure 4: Three pool contrasts from Table 6. Fresh uses a paired-bootstrap 95% CI, pilot the registered-SE two-sided CI, and pooled a one-sided lower bound (open upper cap is display-only). Square area scales with m ; significant heterogeneity makes pooled secondary.

least 8 pp are already plausible on this bed; the “post-selection” label and the bed-local non-claim that follow are calibrated to that scale. Multi-seed replays with pinned, replayable schedule seeds are the follow-up correction (§7). We do not re-run any scoring pass here; both events are recorded properties of the existing $n=819$ archive.

Fresh-pool strata audit. Table 7 re-tabulates frozen per-item contrasts by construction and answer position. The s_conf -over-random margin survives both constructions; the fixed-order advantage retains its sign but is significant only in the answer-pool stratum, and reverses at position 0 (+12.56 pp, $p=0.0031$). The position audit is post-hoc (not pre-registered), but positions 0/1 ($p=0.0031/0.0032$) survive Bonferroni-4 ($\alpha/4=0.0125$), making the reversal multiplicity-controlled rather than selected. These are paired-record re-tabulations, not new decoding.

⁵Non-overlapping beds: the fresh estimate draws on `c1_decisive_20260819-212944` + `c1_fixed_inject_fresh_20260821-062533` ($n=819$), the paired-bed estimate on Trial 1895183 ($n=400$), and the pooled estimate is their sample-size-weighted average ($n=1219$). The three numbers come from non-overlapping beds, different comparators, and different estimands, so we state them separately and do not attribute their difference to any single mechanism.

Table 8: Pooled-to-fresh endpoint decision tree. The final elevation is post-data and is disclosed as such; it is not presented as the original primary endpoint.

Step	Status	Rule or observation	Consequence
1	Pre-data	Pooled $n = 1219$ primary; fresh $n = 819$ sensitivity. Direction fixed as <code>s_conf > rand_inject</code> ; one-sided 95% lower bound must exceed zero. Opposite pooled/fresh signs force UNDETERMINED.	No favorable sign selection
2	Registered data check	Between-pool magnitude heterogeneity: $\delta_{\text{diff}} = 8.8263$ pp, $z = 2.4148$, two-sided $p = 0.0157$; robust $z = 2.3229$.	Reject a common-effect estimate
3	Pre-data branch	Both pool effects are positive (fresh +12.58 pp, pilot +3.75 pp).	Sign-consistency branch passes
4	Post-data	Elevate the disjoint, no-reuse fresh pool; retain pooled +9.68 pp only as a weighted disclosure.	Fresh becomes the secondary answer-level result

Endpoint decision tree. Table 8 separates the pre-data rules from the post-data endpoint elevation. Preregistration did *not* assign fresh-only priority: step 4 is an explicitly post-data choice, motivated by the failure of a common-effect interpretation.

Pool heterogeneity is disclosed, not merged. Figure 4 and Table 6 retain every pool. The fresh comparison is significant under the registered one-sided $\alpha = .05$ rule, but because it was elevated only after pool heterogeneity was observed it is post-selection secondary evidence, pool-dependent and secondary to the $n = 400$ headline; pooled remains only a weighted disclosure. Both signs are positive, and charging two-pool selection still gives a fresh one-sided lower bound of +8.40 pp at $\alpha = 0.025$, while pilot remains undetermined. The pre-specified fresh `s_conf-fixed_inject` adjudication survives both multiplicity accounts (Table 6, note b). This supports a structured-order boundary, not equivalence or a uniquely confidence-driven mechanism.

Boundary: what “reveal order” attributes here, and what it does not. The `s_conf-rand_inject` ablation covers one persisted random schedule. A fixed order that selects its schedule without model output beats `s_conf` on the fresh pool *on average*; within that pool the margin is construction-dependent in strength and reverses at answer position 0 (Table 7). The pooled-fresh sign repeats at $n=400$ but reverses on historical $n=64$ (Figure 5). Because no fixed-order cross-pool test supports merging and Table 8’s heterogeneity concerns `rand_inject`, the former cross-pool mechanism reading is *refuted*. Confidence is neither necessary, uniquely optimal, nor exclusively causal; all six $n=400$ order contrasts remain Holm-nonsignificant (Appendix Table 10). The cleanest same-bed zero-effect row is `s_conf` versus `anti_inject` (+2.00 pp, exact $p=0.573$; Appendix Table 10): reveal direction alone carries neither the reference gap nor the schedule result.

Boundary effect and winner-curse description. Appendix A gives the pilot/extension winner-curse analysis, the $n=400$ undetermined reading, both registered MDE forms, and power rules.

7 LIMITATIONS

Single bed and non-factorial mechanism. One checkpoint and one four-choice GSM8K bed support no cross-model, task, or judge prediction. The `s_conf-mc_uniform` replacement changes feedback, path, and passes jointly, so refuting the narrow compute-artifact account does not factorize it: the reference spends less at one draw but far more at the high-budget read, and the gap survives both there and at measured exact $2L$ (Table 3).

Headline robustness gates, reliability, and seed variation. The frozen rerun, hostile re-adjudication boundary, three-bucket rule, and leave-one-out influence are consolidated in Appendix Table 19: the rerun stays inside its abort tolerance, leave-one-out is a nondegenerate PASS, and the adverse-change counts delimit where significance and direction would fail. None of these supplies the missing multi-seed evidence, and the non-scale-invariant shrink supports only sign and persistence: multi-seed accuracy and τ require new decoding, and the 48-item self- τ is only a noise floor.

Excluded claims and external validity. The rank-half design supports neither dose response nor concavity, and no cross-bed prediction band is used. Under its pre-registered joint zero-and-relevance gate, the five-choice probe remains *UNDETERMINED*: neither supporting evidence nor evidence of absence, and its significant point estimate is not upgraded to external-validity evidence anywhere in this paper (Appendix Table 12). Length-normalized likelihood scorers (acc_norm/NLL) were also not measured; this is a scope limit, not an equivalence or gap. Table 2 localizes only the injected-policy readout.

Efficiency and matched- L boundaries. The scan and forward ledger (Appendix Table 17) cover answer scoring, not generation. The tested curve, including exact $2L$, refutes the narrow artifact but leaves off-curve budget independence open (Table 3). Appendix Table 11 is instead a non-accuracy length identity and no substitute for an accuracy measurement: its nominal all-candidate substitution is a lower audit counterfactual, not input. The three promoted $n=400$ reads all clear the shared 4 pp bar, but the historical high-budget $n = 64$ row remains underpowered (Appendix Table 18).

8 CONCLUSION

On this bed, confidence injection beats the tested non-injected reference, while fixed order wins on average but reverses at position 0 (Table 7). Exact $2L$ refutes only a less-reference-compute account; off-curve, multi-seed, and generalization claims remain unresolved (Appendix H).

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A BOUNDARY ANALYSIS AND REGISTERED VERDICTS

Fresh-pool recipe. The $n = 819$ study uses four-option GSM8K multiple-choice items drawn without question overlap with the $n = 400$ pool; unlike the exactly balanced $n = 400$ pool, its answer positions are only approximately balanced. All arms share the same model checkpoint, prompts, item order, answer extractor, and recorded choice sets. The seeded-random arm is bound to its persisted per-candidate orders because the original runner’s nominal seed derivation included an unpinned Python string hash. The paired record contains one row per question and arm with the prediction, reference answer, correctness, candidate-score vector, and order trace. The anonymous artifact bundle additionally contains the preregistration, execution environment, verdict, and heterogeneity recomputation described in Appendix H.

Table 9: Boundary ledger for `s_conf` versus `rand_inject`, except the directional defence. The original $n = 400$ reveal-order endpoint remains UNDETERMINED; no post-data power calculation changes that label.

Reading	n	$\hat{\Delta}$	m	Exact p	95% CI	Registered reading
First bed	48	+16.67 pp	22	–	[−2.49, +35.82]	Below discordance gate
Extension	64	+7.81 pp	23	–	[−6.87, +22.50]	Below discordance gate
Pilot	400	+3.75 pp	141	0.238	[−2.07, +9.57]	UNDETERMINED
Directional defence	400	+1.75 pp	159	0.634	[−4.43, +7.93]	Wrong sign

As Table 9 shows, the small-bed rows do not distinguish regression to the mean from sampling variation. We use boundary effect only as an analysis-risk description, not as an effect-halving claim. For the $n = 400$ pilot, the interval includes zero and relevant positive effects; the pre-registered terminal reading is thus UNDETERMINED, not equivalence or no effect.

Table 10: All six injected-order contrasts on the $n = 400$ pool. Exact tests are two-sided and Holm-adjusted within this registered family of six (the six pairwise order contrasts are the declared secondary family; the confirmatory $n=400$ budget-curve family of §4 is the separate ten-test Holm family). Each interval is $\hat{\Delta} \pm h$. The last two columns give the two paired-SE conventions: the manuscript quotes the registered $h = 1.96\sqrt{m}/n$ throughout, and the mean-corrected $h = 1.96\sqrt{m - (b - c)^2/n}/n$ differs from it by at most 0.04 pp here (and by at most 1.2 pp on the small $n = 48/64$ pools, where no verdict changes either). Holm adjustment is step-down with monotonicity enforced, so adjusted p is non-decreasing in raw p ; in particular the largest raw p in the family inherits Holm $p = 1.000$.

Pair	m	$\hat{\Delta}$	Exact p	Holm p	Mean-corr. h	Registered h
<code>s_conf-rand_inject</code>	141	+3.75 pp	0.238	0.953	5.807 pp	5.818 pp
<code>s_conf-anti_inject</code>	154	+2.00 pp	0.573	1.000	6.078 pp	6.081 pp
<code>s_conf-fixed_inject</code>	107	−2.75 pp	0.334	1.000	5.061 pp	5.069 pp
<code>rand_inject-anti_inject</code>	159	−1.75 pp	0.634	1.000	6.176 pp	6.179 pp
<code>rand_inject-fixed_inject</code>	138	−6.50 pp	0.033	0.198	5.721 pp	5.756 pp
<code>anti_inject-fixed_inject</code>	153	−4.75 pp	0.145	0.727	6.043 pp	6.061 pp

B LENGTH-IDENTITY CHECK (NO ACCURACY CLAIM)

This section is an arithmetic identity check on candidate lengths. It is not an accuracy measurement and not a substitute for one: the answer-level contrasts of §4 remain the paper’s only accuracy evidence, and their total-NFE-fair $2L/2L$ point is measured directly on the $n=400$ bed. The inputs are the frozen 819-question probe dump and nothing else (per question: candidate lengths $L_i \in \{3, \dots, 9\}$, one designated candidate, and a stored rounding factor ρ_i); no new decoding is involved and no answer-accuracy outcome enters. The distribution the closed form actually consumes is the designated-candidate row of Table 16; the nominal all-candidate distribution is not its input.

Table 11: Length-identity sensitivity (arithmetic projection, no accuracy content). The observed contrast is zero because the probe flag is arm-independent; nonzero cells are closed-form functions of the disclosed designated-candidate lengths. This does not substitute for the measured total-NFE-fair $2L/2L$ row of Table 3. Intervals resample lengths, not accuracy effects. The last row is a lower audit counterfactual based on the nominal all-candidate distribution, not the estimator input.

Restart fraction	0.4	0.6	0.8
Observed-window gap (pp)	0.00	0.00	0.00
Projected gap (pp)	+3.161	+0.692	+0.061
95% CI (pp)	[3.063, 3.252]	[0.657, 0.726]	[0.056, 0.066]
All-candidate substitution (pp)	2.61	0.51	0.04

Definition and closed form. Fix a restart fraction m . Every candidate receives the *same* scalar budget in both arms,

$$\text{NFE}_i(m) = N_i H(L_i) m, \quad N_i = \min(6/L_i, 2),$$

$$H(L) = \frac{1}{(1 - m_L^*)^2}, \quad m_L^* = \frac{1}{1 - 2^{-1/L}},$$

where m_L^* is the reciprocal-scaled form of the half-crossing probability $\tilde{m}_L = 1 - 2^{-1/L}$ that solves $1 - (1 - \tilde{m}_L)^L = \frac{1}{2}$, and H is a currency conversion from restart fraction to forward-evaluation budget rather than a probability ($H(3) = 0.068$ down to $H(9) = 0.006$, mean 0.03). The scale $N_i = \min(c/L_i, 2)$ with $c = 6$ is a design convention: the length-proportional baseline is truncated at 2 so very short candidates receive no steeper allowance, and $c = 6$ keeps $\text{NFE}_i(m) \leq 0.11$ far below the observed decode cost 2^{L_i} , so both service conditions ($2^{L_i} \geq \text{NFE}_i(m)$ for the single deterministic order, $\rho_i 2^{L_i} \geq \text{NFE}_i(m)$ for the random-restart arm) are always satisfied. The random-restart arm crosses to its designated candidate with probability $1 - (1 - m)^{L_i}$, so with ι_i the designated-candidate indicator the per-question gap collapses to

$$\hat{\Delta}_q(m) = \frac{1}{n_c(q)} \sum_{i \in q} [\iota_i - (1 - (1 - m)^{L_i}) \iota_i] = \frac{1}{n_c(q)} \sum_{i \in q} \iota_i (1 - m)^{L_i},$$

and the reported value is its mean over the $Q=819$ questions in percentage points, with a percentile interval from $R=2000$ bootstrap resamples (Efron & Tibshirani, 1994) of the question-level values, seed 20260821.

What the check shows, and its sensitivity. Because the probe records the same designated candidate for both arms, the identification flag is arm-independent: the observed-window contrast is identically 0.00 pp—an algebraic identity, not a null result—and the only inputs that survive are the designated-candidate lengths and m . The counts 163/400/173/48/20/13/2 for $L = 3, \dots, 9$ reproduce the printed 3.161/0.692/0.061 pp values; substituting the nominal all-candidate counts yields the lower 2.61/0.51/0.04 pp audit diagnostic, an assumption *refuted* as the closed-form input. Replacing $c = 6$ by $c = 4$ leaves every cell of Table 11 bit-identical, because the gap formula $\iota_i (1 - m)^{L_i}$ never touches N_i or H ; and relabeling the two arms or randomizing all recorded correctness outcomes leaves the projected column unchanged while the observed-window row stays identically zero, confirming that the projected column carries no outcome information at all. The dominant assumption is therefore the restart model itself—independent restarts with a uniform designated-target probability m , a stylized model of a random-restart scorer and not a measured property of either arm. The projected column is a length-distribution identity of that model, printed for budget-accounting transparency; it is not evidence that a compute-fair accuracy gap would shrink, grow, or persist, and it can neither confirm nor refute §4.

C FIVE-OPTION EXTERNAL-VALIDITY PROBE

Table 12: Five-option probe. The terminal label follows the pre-registered rule and is not reinterpreted after observing power. The significant point estimate below therefore carries no evidence-of-absence interpretation and no external-validity claim; it is not cited as a measured conclusion anywhere outside this table and the matching limitation in §7.

Metric	Value
Items decoded	400/400
s_conf accuracy	62.25%
rand_inject accuracy	53.75%
Δ	+8.5 pp
Discordant b/c	84 / 50
One-sided exact p	0.0021
One-sided 95% lower bound	+3.74 pp
Planning point	7.59 pp at $m = 149$
Realized discordance	$m = 134$
Registered verdict	UNDETERMINED

The realized power calculation is descriptive. It neither replaces the registered discordance gate nor upgrades this result to supporting evidence.

D REGISTRATION AND ANALYSIS CHRONOLOGY

Table 13: Anonymous chronology of the load-bearing decisions. Pre-data and post-data steps are kept distinct.

Status	Decision	Consequence
Pre-data	Pooled $n = 1219$ primary; fresh $n = 819$ sensitivity; opposite signs force UNDETERMINED.	Prevents choosing a favorable sign.
Pre-data	One-sided direction and heterogeneity check fixed before the fresh outcome.	Separates direction from pool compatibility.
Registered data check	Fresh and reused-pilot magnitudes differ ($z = 2.4148, p = 0.0157$).	Common-effect pooling is rejected.
Post-data, disclosed	Fresh disjoint pool elevated; pooled estimate retained as secondary disclosure.	The fresh priority is not presented as original.
Pre-data extension	Fixed identity order compared with s_conf on the fresh pool as a single load-bearing pair.	Tests whether confidence ordering is necessary or uniquely optimal.
Post-data reading	Fixed order beats s_conf.	Supports only a structured order boundary, with format/structure caveats.

E HISTORICAL DIAGNOSTICS AND BUDGET LEDGER

Table 14: Historical $n = 64$ estimator-budget scan (upper rows, exploratory) and the corrected $n = 400$ confirmatory scan (lower block, ascending budget). The historical $n_{mc} = 128$ row remains underpowered under the separate 25 pp exploratory tier. On the $n = 400$ bed, the pre-registered draw-matched $n_{mc,i} = L_i$, total-NFE-fair $n_{mc,i} = 2L_i$, and high-budget $n_{mc} = 128$ contrasts share the same 4 pp promotion bar and all three pass; the last is the conservative main read. Intervals follow Table 3: question bootstrap for the two $n_{mc,i}$ rows, registered paired SE elsewhere.

Reference	Δ	m	Exact p	95% CI
Single draw ($n=64$)	+25.00 pp	30	0.0052	[8.23, 41.77]
$n_{mc} = 128$ ($n=64$)	+10.94 pp	29	0.2649	[-5.56, +27.43]
Per-candidate draw-matched ($n=64$)	+25.00 pp	32	0.0070	[7.68, 42.32]
<i>Corrected $n = 400$ confirmatory run</i>				
Single draw ($n=400$)	+14.00 pp	170	2.1×10^{-5}	[7.61, 20.39]
Draw-matched $n_{mc,i} = L_i$ ($n=400$)	+13.00 pp	160	4.8×10^{-5}	[7.00, 19.00]
Total-NFE-fair $n_{mc,i} = 2L_i$ ($n=400$)	+13.00 pp	166	6.6×10^{-5}	[6.75, 19.25]
$n_{mc} = 128$ ($n=400$)	+9.00 pp	128	0.0019	[3.46, 14.54]

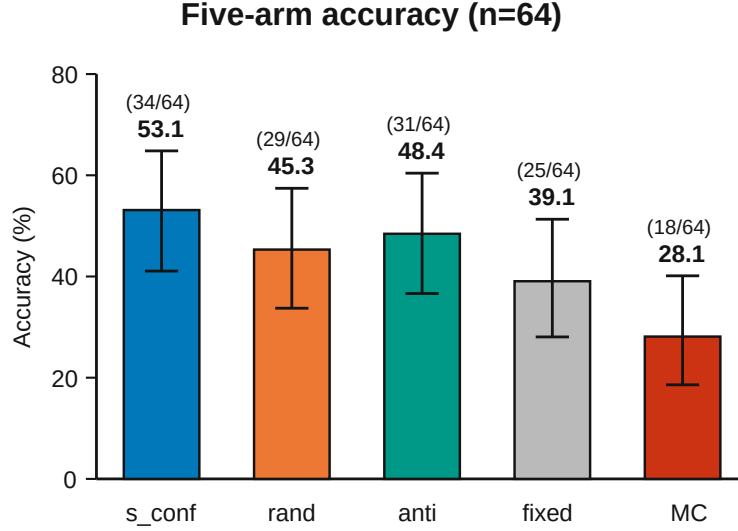


Figure 5: Historical five-arm top-1 accuracy on the $n = 64$ pool, regenerated from frozen item-level records; exact counts are printed above each bar. Whiskers are descriptive Wilson 95% intervals (Wilson, 1927) on each arm’s *marginal* accuracy. They overlap heavily at this sample size, which is why this pool is exploratory only; the registered comparisons are paired and are reported in Table 14, not read off these unpaired intervals.

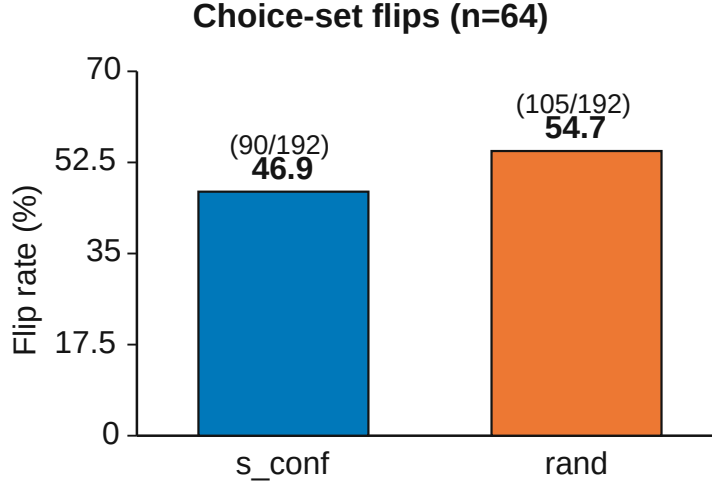


Figure 6: Descriptive diagnostic: flip rates under a uniform 25% distractor-score shrink ($n=64$ panel; the $n=48$ panel is sign-consistent and omitted for space). Exact counts are printed with the bars; the shrink is a proportional rescaling on each protocol’s own score scale, is not scale-invariant across scorers, and no inferential interval is implied.

Table 15: Rank-agreement diagnostics. Deterministic rows are reproducibility anchors; random-order self-agreement is the stochastic comparator floor.

Diagnostic	Scope	Value	Interpretation
s_conf-rand_inject tie-drop	$n = 48$	0.2514	Full grid 0.2500; 8 tied items
s_conf-rand_inject tie-drop	$n = 64$	0.3323	Full grid 0.3255; 9 tied items
Random-order self- τ	Shared 48	0.2361 ± 0.451	Stochastic floor
Deterministic self- τ	Shared 48	1.000	Reproducibility anchor
s_conf-fixed_inject	$n = 400/\text{fresh } 819$	$0.48 / 0.47$	Cross-protocol anchor

Table 16: Candidate-length distributions used by draw matching and strict NFE sensitivity. The first three rows count all four nominal candidates per question. The final row counts only the unique designated candidate in each of the 819 fresh questions; this is the distribution used by the closed form in Table 11.

Pool	Items	$L = 3$	$L = 4$	$L = 5$	$L = 6$	$L = 7$	$L = 8$	$L = 9$	Mean L
Historical	64	39	120	72	17	8	0	0	4.355
Primary	400	262	751	424	118	30	15	0	4.343
Fresh sensitivity	819	364	1237	1010	390	167	101	7	4.722
Fresh designated	819	163	400	173	48	20	13	2	4.278

Table 17: Per-candidate budget ledger, in ascending reference budget. Scoring draws and total forward calls are deliberately separated; schedule selection adds another L forwards to the confidence and anti-confidence arms. The reference-arm rows carry the same budget labels as Table 3, so every row of that budget curve closes back to a ledger line: single draw, $n_{mc,i} = L_i$ (nominal draw match only), $n_{mc,i} = 2L_i$ (total-NFE match), and $n_{mc} = 128$.

Protocol	Scoring draws	Selection forwards	Total forwards	Reading
s_conf /	L	L	$2L$	Model-derived ordering plus injection
anti_inject				
rand_inject /	L	0	L	Seeded-random or fixed ordering plus injection
fixed_inject				
mc_uniform	1	0	1	One masked-subset estimator draw
single				
mc_uniform	L	0	L	Matches nominal injection scoring draws only
draw-matched				
$n_{mc,i} = L_i$				
mc_uniform	$2L$	0	$2L$	Matched to s_conf’s total NFE (scoring plus selection); the decisive row of Table 3
total-NFE-fair				
$n_{mc,i} = 2L_i$				
mc_uniform high budget	128	0	128	High-budget point on the historical pilot and on the corrected $n=400$ run

F CONTRIBUTION AND POWER DETAILS

Table 18: Answer-accuracy contribution ledger. The $n = 400$ total-NFE-fair $2L/2L$ row is the decisive equalization and the draw-matched $n_{mc,i} = L_i$ row is the pre-registered primary endpoint; the four $n = 400$ rows form the reported budget curve and the high-budget endpoint is the conservative main read. “Per-candidate draw-matched” sets the reference estimator budget at $n_{mc,i} = L_i$ per candidate. The $n = 64$ row alone retains the separate historical 25 pp exploratory tier.

Reference point	n	$\hat{\Delta}$	m	Exact p	95% CI	Role
Single draw	400	+14.0 pp	170	2.1×10^{-5}	[7.61, 20.39]	Curve point
Draw-matched $n_{mc,i} = L_i$	400	+13.0 pp	160	4.8×10^{-5}	[7.00, 19.00]	Pre-registered primary
Total-NFE-fair $n_{mc,i} = 2L_i$	400	+13.0 pp	166	6.6×10^{-5}	[6.75, 19.25]	NFE-fair decisive
$n_{mc} = 128$	400	+9.0 pp	128	0.0019	[3.46, 14.54]	Conservative main read
Per-candidate draw-matched	64	+25.0 pp	32	0.007	[7.68, 42.32]	Exploratory (25pp tier)

G POWER AND SAMPLE-SIZE LEDGER

The decisive total-NFE-fair $2L/2L$ contrast clears the registered Holm threshold and shared promotion bar; the historical small-pool draw-matched result is detected with only a slim margin over its detectable magnitude, while the historical small-pool high-budget arm is underpowered and remains UNDETERMINED (not zero) under its exploratory tier. The corrected $n = 400$ high-budget read is determined and clears the same promotion bar as the decisive $2L/2L$ row (Table 18). Rank agreement and fixed perturbation overlays are descriptive diagnostics rather than hypothesis tests.

Table 19: Headline robustness-gate ledger. The frozen rerun is within its abort tolerance, and the leave-one-out check is a nondegenerate PASS; the hostile re-adjudication counts state the boundary at which significance or direction would fail.

Diagnostic	Recorded quantity	Reading
Frozen rerun	$n=400$; s_{conf} accuracy 42.75%; deviation 0.0 pp; abort tolerance 1 pp	Within abort tolerance
Hostile $b \rightarrow c$ re-adjudication	166 discordants; one change moves the discordant proportion by 0.60 pp and $\hat{\Delta}$ by $2/n=0.50$ pp; 12 changes lose significance and 26 reverse the sign	Change-count boundary
Three-bucket rule	$0 < \text{infl} \leq \text{margin}$: PASS; $\text{infl} > 2/n$: FAIL; zero influence: degenerate	Prespecified categorization
Leave-one-out	Influence 0.283 pp on the 400-item bed; $1/n=0.25$ pp; margin/influence ≈ 21	Nondegenerate PASS

H ANONYMOUS REPRODUCTION CHECKLIST

Table 20: Components included in the anonymous reproduction bundle. Each row has an immutable manifest entry and a one-command validation check.

Component	Contents	Independent check
Sampling frames	Question identifiers, split membership, option count, and reference-option index for every pool	Verify sample sizes, disjointness, and four-versus five-option chance levels
Distractor construction	Deterministic rules for generating, deduplicating, and shuffling numerical distractors	Rebuild each choice set and compare all candidate strings
Prompt template	Exact instruction, question, choices, answer suffix, and mask placement	Byte-compare rendered prompts before decoding
Protocol implementation	Confidence, anti-confidence, seeded-random, fixed, and random-mask schedules; tie policy; seed policy and persisted order traces	Run sampler self-checks, replay recorded schedules, and detect the documented cross-process seed limitation
Judge and extraction	Top-1 argmin rule, tie handling, answer parser, and invalid-output policy; no partial credit	Recompute predictions and paired correctness from score vectors
Statistics	Exact McNemar, paired intervals, Holm families, binomial chance tests, heterogeneity check, and SESOI formula	Recompute every table cell from the item-level records
Figures	Deterministic regeneration scripts plus the plotted data payloads they emit	Regenerate quantitative PDFs and byte-compare their numerical payloads
Manifest	Content hashes, environment specification, and command sequence	Validate dependencies before building the canonical manuscript

Table 20 is the public traceability surface. Internal workflow labels, machine paths, task identifiers, and revision tokens are intentionally absent from the double-blind manuscript while remaining available to the artifact auditor.